

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

C02: *Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I _____ V.Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Scenario-Based

1. Smart Campus Subnetting (192.168.100.0/24)

A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Answer:

1. Network Address and Broadcast Address:

- School of Computing (/25): Network address = 192.168.100.0, Broadcast address = 192.168.100.127
- School of Electronics (/26): Network address = 192.168.100.128, Broadcast address = 192.168.100.191
- School of Mechanical Engineering (/27): Network address = 192.168.100.192, Broadcast address = 192.168.100.223

2. Valid Host IP Address Range:

- School of Computing: 192.168.100.1 – 192.168.100.126
- School of Electronics: 192.168.100.129 – 192.168.100.190
- School of Mechanical Engineering: 192.168.100.193 – 192.168.100.222

3. Number of Usable Host Addresses:

- School of Computing (/25): $2^7 - 2 = 126$ usable hosts
- School of Electronics (/26): $2^6 - 2 = 62$ usable hosts
- School of Mechanical Engineering (/27): $2^5 - 2 = 30$ usable hosts

4. Remaining Unused IP Address Range:

- After allocating the three subnets (192.168.100.0 – 192.168.100.223), the block 192.168.100.224 – 192.168.100.255 (a /27, 30 usable addresses) remains unused and available for future expansion.

2. Multinational Company Subnetting (10.10.0.0/16)

A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid range of host IP addresses.
3. Calculate the number of usable hosts in each subnet.
4. Identify the unused IP address range remaining within the allocated IP block.

Answer:

1. Network Address and Broadcast Address:

- Software Development (/24): Network = 10.10.0.0, Broadcast = 10.10.0.255
- Quality Assurance (/25): Network = 10.10.1.0, Broadcast = 10.10.1.127
- Human Resources (/26): Network = 10.10.1.128, Broadcast = 10.10.1.191
- Executive Management (/27): Network = 10.10.1.192, Broadcast = 10.10.1.223

2. Valid Range of Host IP Addresses:

- Software Development: 10.10.0.1 – 10.10.0.254
- Quality Assurance: 10.10.1.1 – 10.10.1.126
- Human Resources: 10.10.1.129 – 10.10.1.190
- Executive Management: 10.10.1.193 – 10.10.1.222

3. Number of Usable Hosts:

- Software Development (/24): $2^8 - 2 = 254$ usable hosts
- Quality Assurance (/25): $2^7 - 2 = 126$ usable hosts
- Human Resources (/26): $2^6 - 2 = 62$ usable hosts
- Executive Management (/27): $2^5 - 2 = 30$ usable hosts

4. Unused IP Address Range Remaining:

- After allocating all four departments (10.10.0.0 – 10.10.1.223), the range 10.10.1.224 – 10.10.255.255 remains unused/unallocated within the 10.10.0.0/16 block for future departments.

3. IPv4 to IPv6 Migration (2001:0db8:abcd:0000:0000:0000:0000/64)

A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its compressed notation.
2. Expand the compressed address back into its full hexadecimal representation.
3. Identify the network prefix using the /64 prefix length.
4. Calculate the total number of host addresses supported within the subnet.
5. Calculate the total number of possible host addresses available within this network.

Answer:

1. Compressed Notation:

- 2001:db8:abcd::/64 (consecutive groups of zeros are replaced with :: and leading zeros in each group are dropped)

2. Expanded (Full Hexadecimal) Form:

- 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64):

- 2001:0db8:abcd:0000 — the first 64 bits (first four hexets) identify the network; the remaining 64 bits form the interface ID.

4. Total Host Addresses Supported in the /64 Subnet:

- $2^{(128-64)} = 2^{64} = 18,446,744,073,709,551,616$ addresses

5. Total Possible Host Addresses Available Within This Network:

- Since /64 leaves all 64 interface-ID bits free, the total possible host addresses = $2^{64} \approx 1.8 \times 10^{19}$ addresses (the same value as above).

4. ISP Routing Table Lookup

An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address: 192.168.2.45

As the routing engineer, determine:

1. The destination subnet to which the packet belongs.
2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

Answer:

1. Destination Subnet:

- 192.168.2.45 lies within the range 192.168.2.0 – 192.168.2.255, so the packet belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry:

- 192.168.2.0/24 — this is the only entry whose network range contains the destination address 192.168.2.45.

3. Next-Hop Network/Interface:

- The router forwards the packet out of the interface directly connected to the 192.168.2.0/24 network (the LAN segment serving that subnet).

4. Default Route (if no match is found):

- If the destination address does not match any entry in the table, the router uses the default route 0.0.0.0/0 (the gateway of last resort) to forward the packet.

5. Two-Department LAN via Layer-3 Router

A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each LAN.
2. Identify the valid host IP address range.
3. Suggest suitable IP addresses for three computers in each LAN.
4. Identify the default gateway that hosts in each LAN should use for communication.

Answer:

1. Network Address and Broadcast Address:

- Administration LAN (192.168.10.0/26): Network = 192.168.10.0, Broadcast = 192.168.10.63
- Finance LAN (192.168.10.64/26): Network = 192.168.10.64, Broadcast = 192.168.10.127

2. Valid Host IP Address Range:

- Administration LAN: 192.168.10.1 – 192.168.10.62

- Finance LAN: 192.168.10.65 – 192.168.10.126

3. Suggested IP Addresses for Three Computers:

- Administration LAN: 192.168.10.2, 192.168.10.3, 192.168.10.4
- Finance LAN: 192.168.10.66, 192.168.10.67, 192.168.10.68

4. Default Gateway for Each LAN:

- Administration LAN hosts use gateway: 192.168.10.1
- Finance LAN hosts use gateway: 192.168.10.65

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand